Zeros from the radical:
MUSIC on the truncated Weil form, hyper-nullity, and the precision law

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Abstract

Let Q be the truncated Weil quadratic form of ζ (or of a Dirichlet L-function) on the N-dimensional even window space at $L = \log\mu$, assembled from the pole, the archimedean term and the primes $\le \mu$ — no zeros are used. We show that the low end of its spectrum (the *radical*) is a rigorous noise subspace in the sense of MUSIC: if $Q(v) = \lambda$ then $|\hat{v}(\gamma)| \le \sqrt{\lambda/2}$ at every zero γ. Locating the minima of $\|P_{\mathrm{rad}}\hat{c}(\gamma)\|$ recovers the zeros of ζ to 10^{-19} from a 47×47 matrix built from five primes, and $\gamma_1(\chi_3)$ to 4×10^{-58} at $\mu = 38$. We prove that the sequence of transforms of the bottom vector at the zeros is itself the bottom eigenvector of the zero Gram (Proposition A), measure the precision law $\mathrm{err}(\gamma_1) \approx e^{-s(\chi)\mu}$ — the drilling speed is the per-digit cost of reading the first zero — and trace it to an endpoint law $|\hat{v}_0(\gamma_1)| \approx C\lambda_0$, $C \in [7, 28]$ over sixty orders of magnitude, whose mechanism (the radical's mass is expelled to the band edge) we quantify. We also record the certified positive definiteness of the complete form at $\mu = 11$, which gives the quorum theorem its second half. Everything is a statement about the explicitly defined forms; nothing bears on RH.

1 Setting and the identity

$V_N = \mathrm{span}(\eta_0, \dots, \eta_{N-1})$, $\eta_n(x) = \sqrt{2/L}\cos(\omega_n(x + L/2))$ on $[-L/2, L/2]$, $\omega_n = 2\pi n/L$ ($\eta_0 = 1/\sqrt{L}$). Q is the prime-side matrix $Q_{nm} = P_{nm} + A_{nm} - \sum_{p^k\le\mu} \frac{\log p}{p^{k/2}}\Theta_{nm}(k\log p)$ (notebook §15, note *quorum-theorem* App. B). The zero evaluator is $\hat{c}(\gamma)_n = \hat{\eta}_n(\gamma)$, in closed form

$$\hat{\eta}_0(\gamma) = \frac{2\sin(\gamma L/2)}{\gamma\sqrt{L}}, \qquad \hat{\eta}_n(\gamma) = \sqrt{\tfrac{2}{L}}\,\sin(\gamma L/2)\,\frac{2\gamma}{\gamma^2 - \omega_n^2} \quad (n \ge 1),$$

a Dirichlet kernel. Under RH the explicit formula reads $Q = \sum_{\gamma>0} 2\,\hat{c}(\gamma)\hat{c}(\gamma)^\top$; we verified it as a matrix identity (residual 8.9% after the 40 zeros below the band edge $\omega_{\max} = 120.5$ at $\mu = 11$, then an algebraic $1/G$ tail; closed to $O(1/G^2)$ on V in the note *squares47*).

2 The radical is a noise subspace

Proposition 1 (admission bound)**.** *If $Q(v) = \lambda$ and $\|v\| = 1$ then $|\hat{v}(\gamma)| \le \sqrt{\lambda/2}$ for every zero γ (under RH; unconditionally for the zeros on the line up to the verified height, with the usual tail).*

Proof. $\lambda = \sum_\gamma 2\hat{v}(\gamma)^2 \ge 2\hat{v}(\gamma_k)^2$. □

Hence the d deepest eigenvectors span a subspace on which every zero evaluator is small, and the MUSIC pseudo-spectrum $\|\hat{c}(\gamma)\|^2/\|P_{\mathrm{rad}}\hat{c}(\gamma)\|^2$ peaks at the zeros. The bound is the *admission criterion*: a vector with $\lambda = 0.12$ admitted into the noise subspace shifts every peak by $+8$ (notebook §16.2); with $\lambda \le 5.6\times 10^{-10}$ the same character's zeros come out to 10^{-9}.

form	μ	noise rungs λ	zeros recovered	precision on γ_1
ζ	11	3 rungs $\leq 2 \times 10^{-34}$	first 12; 30/40 in band	9×10^{-20}
$L(\cdot, \chi_3)$	11	2 rungs $\leq 5.6 \times 10^{-10}$	11/48 in band	2.5×10^{-9}
$L(\cdot, \chi_3)$	38	1 rung 7.3×10^{-61}	—	4.16×10^{-58}

The $\mu = 38$ value is checked against an independent 60-digit Hurwitz computation of $\gamma_1(\chi_3) = 8.039737155681466681713623214172965802793010267386061427270 9\ldots$; the localization mechanism closes exactly: err $= |\hat{v}_0(\gamma_1)|/|\hat{v}_0'(\gamma_1)| = 6.4 \times 10^{-60}/0.0154 = 4.16 \times 10^{-58}$.

3 The Gram duality

Proposition 2 (A)**.** *Let $Qv_0 = \lambda_0 v_0$ and $\hat{v} = (\hat{v}_0(\gamma_k))_k$. Then $G\hat{v} = (\lambda_0/2)\hat{v}$, where $G_{jk} = \langle \hat{c}(\gamma_j), \hat{c}(\gamma_k)\rangle$ is the (infinite) Gram matrix of the zero evaluators.*

Proof. From $Qv_0 = \lambda_0 v_0$ and $Q = \sum_k 2\hat{c}_k\hat{c}_k^\top$: $v_0 = (2/\lambda_0)\sum_k \hat{v}_k\hat{c}_k$; pair with $\hat{c}_j$. □

The radical in function space and the quasi-kernel of the zero Gram are the same object. The identity cannot be checked by truncation: on 45 zeros $[G\hat{v}]_1 \approx 10^{-28}$ against a target $(\lambda_0/2)\hat{v}_1 \approx 10^{-94}$ — the zeros beyond the 45th cancel the in-band Gram action to 66 digits (98 at $\mu = 16$). It is established indirectly by the eigen-residual $\|Qv_0 - \lambda_0 v_0\| = 10^{-61}$.

4 Envelope versus individuals

At $N = 47$ the zero evaluators live entirely in the complement of the radical (mass 1.000000 for every in-band zero), but no top eigenvector is an individual zero: overlaps 0.50–0.93 with near-equal seconds, driven by the coherence of neighbouring Dirichlet kernels (spacing ≈ 2.3 against width ≈ 2.6). At small N (13–21, few zeros in band) the top directions *are* individual evaluators (cosines $0.90, 0.84, 0.84$; one-to-one matching up to $K = 6$, breaking at the last in-band zero, notebook §25, §31, §53). Individuality lives on the band core and in the noise subspace, not in the signal squares at large N.

5 Precision law, endpoint law, leakage

Precision. Between $\mu = 11$ and 38 for χ_3, $-\ln \mathrm{err}(\gamma_1)$ grows from 19.8 to 132.1: slope $4.16 \approx s(\chi_3) = 4.00$. The error follows $e^{-s\mu}$, full depth, not $e^{-s\mu/2}$: *the drilling speed is the per-digit cost of reading the first zero.*

Endpoint law. $|\hat{v}_0(\gamma_1)| \approx C\lambda_0$ with $C = 22$ ($\zeta, \mu = 11$), 24 ($\zeta, \mu = 16$), 27.8 (ζ, $N = 21$), 8.8 ($\chi_3, \mu = 38$), 7 ($\chi_3, \mu = 11$): proportional to λ_0, not to the admission bound $\sqrt{\lambda_0/2}$ — at $\mu = 38$, $|\hat{v}_0(\gamma_1)| = 6.4 \times 10^{-60}$ against a bound 6×10^{-31}. This *hyper-nullity* is the single root of the precision law (err $= |\hat{v}|/|\hat{v}'| \sim \lambda$) and of the leakage profile below.

Leakage spectrum. $|\hat{v}_0(\gamma_k)|$ measured on 45 zeros of ζ at 85 digits rises from 7.8×10^{-47} at γ_1 to a peak 5.3×10^{-25} at $\gamma = 124.3$, just past the band edge 120.5; the 55 first zeros carry only 42% of λ_0, the rest is an algebraic out-of-band tail. Inside the band $\ln|\hat{v}_0(\gamma)| \approx -\tau(\omega_{\max} - \gamma)$ with a mildly decreasing local rate ($0.64 \to 0.49$ at $\mu = 16$); τ is basis-independent, grows mildly with μ, depends on the L-function (0.48 for ζ, 0.27 for χ_3 at $\mu = 11$), and satisfies $\tau \approx s\mu/(2\omega_{\max})$ to 6% on ζ: the profile interpolates from $\sim \lambda_0$ at γ_1 to $\sim \sqrt{\lambda_0}$ at the edge. *The difficulty of truncated positivity is localized in an $O(1/\tau)$ neighbourhood of the band edge.* Two artifacts caught along the way are recorded in the notebook (zeros converted before setting the precision — a floor at slope$\times 10^{-16}$ identical across configurations; a fourth "noise" vector with $\lambda = 0.12$).

6 The other half of the quorum theorem

The note *quorum-theorem* certifies that every proper sub-Euler product violates positivity at $\mu = 11$. The complete form is now certified positive definite there: direct Cholesky in balls dies at pivot 14 (condition number $\sim 10^{48}$), but $M = V^\top QV$ with V a *floating* eigenbasis (its accuracy is irrelevant to rigour), Q in Arb balls with radii $\leq 10^{-55}$, is certified strictly diagonally dominant: critical row $M_{00} = 3.58317 \times 10^{-48} \pm 3 \times 10^{-54}$ against off-diagonal sum $\leq 8.5 \times 10^{-54}$; hence $M \succ 0$, V invertible, $Q \succ 0$ (Sylvester). 152 s, `code/positivite_certifiee.py`. Both halves of the theorem exist at $\mu = 11$: every proper sub-product violates, the complete product is positive definite, landing at 3.58×10^{-48} from zero with certified error bars.

Status and reproducibility

Propositions 1 and A are proved; the certification is a computation with rigorous error bounds; everything else is measurement with stated numbers. Scripts: `code/music_zeros.py`, `code/positivite_certif` zero caches `code/zeros_zeta_90_hp.pkl`, `zeros_chi3_hp.pkl`. The recovery of zeros from the truncated form is Groskin's superexponential convergence phenomenon; our contribution is the lens — the admission bound, the subspace formulation, the Gram duality, and the laws that tie precision to depth.